

# The Hardy Cell Dataset: Measurement Philosophy, Protocol, and Results

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10,000 Riemann zeta zero intervals on the critical line — certified under frozen Protocol v2

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*This publication presents a complete dataset of 10,000 Hardy cells on the critical line ( $s = 1/2 + it$ ), measured under a frozen, preregistered protocol designed to eliminate retroactive fitting and model adjustment. A Hardy cell is the interval between two consecutive nontrivial Riemann zeta zeros. Within each cell, two quantities were measured: the interior transverse breadth of Hardy's Z-function, and the normalized location of that breadth's maximum. The work combines three elements: (1) an epistemological foundation establishing why protocol freezing before data acquisition is essential for mechanical honesty, (2) the complete specification of Protocol v2, including five independent measurement gates with explicit numerical tolerances, and (3) the full dataset with 33 measured and derived fields per cell. We document the certification pathway and the convergence diagnostics that establish measurement stability. This dataset stands alone and requires no external framework to interpret — its value lies in the reproducibility guarantee and the integrity of the measurement boundary itself.*

## What Are Hardy Cells?

A Hardy cell is the interval between two consecutive nontrivial zeros of the Riemann zeta function  $\zeta(s)$  on the critical line  $\text{Re}(s) = 1/2$ . For zeros  $\gamma_n$  and  $\gamma_{n+1}$ , the cell is the open interval  $(\gamma_n, \gamma_{n+1})$ , of width  $h_n = \gamma_{n+1} - \gamma_n$ . Within this interval, two quantities are measured:

- The transverse breadth of the Hardy Z-function,  $W_{n\perp}$  — twice the maximum of  $|Z(t)|$  over the cell's interior.
- The normalized location  $x_n$  of that interior maximum within the cell, where  $x = 0$  is the lower boundary,  $x = 1$  the upper boundary, and  $x = 0.5$  the midpoint.

Hardy cells are not imposed by external theory — the interval itself is a structural fact about the zeros. Every Riemann zero lies at the boundary of two Hardy cells. This dataset measures 10,000 such cells using a frozen protocol, establishing that the measurement of breadth and location is stable across the

two independently specified resolution paths used by Protocol v2, demonstrating robustness when the method is locked before data acquisition.

# 1. Epistemological Foundation

## 1.1 The Core Principle: Deterministic Observation as Ground Truth

This work rests on a single epistemological commitment: deterministic observation is the ground truth standard. When we measure a mathematical object, the measurement result is what it is — a fact about the object, not a hypothesis to be refined by post-hoc adjustment.

The burden of explanation for any discrepancy between observation and theory lies entirely with the theory, not with the observation. If a model predicts a given value for a measured quantity but the certified data shows otherwise, the model must be revised. We do not adjust the observation or dilute it with auxiliary hypotheses.

This principle is enforced methodologically through preregistration and protocol freezing. By locking the measurement specification, decision rules, and numerical thresholds before data acquisition, we eliminate researcher degrees of freedom — the freedom to retroactively soften a gate because results are "unexpected," to adjust a threshold to achieve a desired outcome, or to selectively analyze a subset of the data. The theory must now explain the data as it was measured, not the other way around.

## 1.2 Protocol Freezing: Why It Matters

A frozen protocol is a specification locked before data acquisition. Every measurement gate, every tolerance, every numerical threshold is written down and sealed. No parameters are adjusted after data arrives. No gates are softened because early results are "outliers." No decision boundaries are moved to achieve a desired outcome.

The purpose is not paranoia. It is mechanics: once you allow retroactive adjustment, you have admitted a hidden degree of freedom. Each cell measurement can then be "explained" by tuning a parameter per cell. The model appears to fit perfectly — but you have fit the noise, not the signal.

Protocol freezing removes that freedom. It forces the model to make hard claims: "This is what I predict will happen when you measure 10,000 independent cells. Here are the exact gates I will use to judge success or failure."

The result is binary: PASS or FAIL. No partial credit. No adjustment. If the model fails, it fails publicly and completely. That failure is more valuable than a thousand post-hoc explanations, because it is honest.

## 1.3 The Hypothesis Seal

Protocol v2 was frozen on September 2, 2026, before the certified measurement run. During acquisition, analysis of the following was explicitly forbidden: distributions of the normalized location  $x_n$ , breadth-to-width correlations, "breathing" or cyclical pattern language, height evolution, predictive or causal relationships, or mining for geometric patterns. This firewall ensured the dataset captured what was measured, not what was expected.

## 1.4 Intellectual Honesty

In an era when it has become trivially easy to overfit, and when publication incentives reward surprising results, frozen-protocol datasets are a hedge against self-deception. This dataset makes an implicit claim: we were willing to be wrong. We locked in our measurement method before we knew what the answer would be. If a prediction against this data fails, the failure is real and not rationalized away. That willingness to be falsifiable is the dataset's deepest value.

# 2. Protocol v2: Complete Measurement Specification

## 2.1 Overview

Protocol v2 defines five independent measurement gates, each with explicit pass/fail criteria. A cell passes certification only if it clears all five gates.

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Gate 1: Breadth Solver (Derivative-Free)
Gate 2: Interior Maximum Boundary Check
Gate 3: Diagnostic Transparency
Gate 4: Refinement Stability ( $\epsilon$  tolerance)
Gate 5: Raw Integrity (Provenance)
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## 2.2 Measurement Procedure

Each cell was measured independently at two resolutions to verify convergence:

- Resolution A: 64-point grid, 28 decimal places of precision.
- Resolution B: 128-point grid, 40 decimal places of precision (final reported value).

Candidates for the interior maximum were located by direct grid scanning — no derivative information — then bracketed and refined with 32 iterations of golden-section search. No  $Z'(t)$  information entered the breadth measurement at any stage.

## 2.3 Gate 1: Breadth Solver (Derivative-Free)

**Purpose:** Verify that the frozen derivative-free algorithm successfully locates the cell's interior

maximum and computes breadth.

**Criterion:** The algorithm must identify exactly one interior maximum (not at a boundary) at both Resolution A and Resolution B, with  $W_A > 0$  and  $W_B > 0$ .

**Status across dataset:** **10,000 / 10,000 PASS (100%)**

## 2.4 Gate 2: Interior Maximum Boundary Check

**Purpose:** Verify that the detected maximum lies strictly inside the cell boundaries, not at or outside them.

**Criterion:**  $\gamma_n < t_{\max} < \gamma_{n+1}$  at both resolutions.

**Status across dataset:** **10,000 / 10,000 PASS (100%)**

## 2.5 Gate 3: Diagnostic Transparency

**Purpose:** Verify that full diagnostic information is recorded for independent audit, and that stationary-root status is explicitly disclosed rather than silently assumed.

**Criterion:** Stationary-root diagnostic status must be explicitly recorded. The status UNRESOLVED is a permitted, passing value — stationary-root success is not required for breadth certification.

**Status across dataset:** **10,000 / 10,000 PASS (100%)**. Stationary-root status is recorded as UNRESOLVED for 10,000 / 10,000 cells: this diagnostic was not attempted in this run, by design.

## 2.6 Gate 4: Refinement Stability (Convergence Test)

**Purpose:** Verify that measurements are numerically stable and represent true cell properties, not grid artifacts, by requiring tight agreement between the two independent resolutions.

$$\begin{aligned}\varepsilon_W &= |W_B - W_A| / \max(|W_A|, |W_B|) && \text{threshold: } < 10^{-9} \\ \varepsilon_x &= |x_B - x_A| && \text{threshold: } < 10^{-5}\end{aligned}$$

**Status across dataset:** **10,000 / 10,000 PASS (100%)**. Observed  $\varepsilon_W$  ranged from  $1.042 \times 10^{-30}$  to  $2.626 \times 10^{-15}$ ; observed  $\varepsilon_x$  ranged from  $5.441 \times 10^{-27}$  to  $8.957 \times 10^{-8}$ . Both extremes remain below their respective thresholds.

## 2.7 Gate 5: Raw Integrity (Provenance and Protocol Adherence)

**Purpose:** Verify that the measurement record is genuine — measured under the preregistered protocol, from the preregistered data source, with no alteration to the measurement pipeline.

**Criterion:** Zero boundaries obtained from the designated LMFDB source file, cell index contiguous and correct, Protocol v2 applied without modification, no smoothing, repair, substitution, or alteration to raw measurements.

**Status across dataset:** 10,000 / 10,000 PASS (100%)

## 2.8 Certification Summary

Status = PASS ⇔ (Gate1 AND Gate2 AND Gate3 AND Gate4 AND Gate5)  
Status = REJECT if any gate fails

No retests. No partial passes. All 10,000 cells received all\_five\_gates = PASS.

## 3. Dataset Acquisition and Certification

Protocol v2 was frozen on September 2, 2026. The measurement run covers cells 1 through 10,000, drawing zero boundaries  $\gamma_n$  from the designated LMFDB source. Total cells certified: 10,000. Pass rate: 100% (10,000 / 10,000). No q-space or hypothesis inspection occurred during acquisition; the hypothesis seal (Section 1.3) remained in force throughout.

## 4. Results

### 4.1 Certification Outcomes

| Quantity                              | Value                    |
|---------------------------------------|--------------------------|
| Total cells acquired                  | 10,000                   |
| Cells certified (all five gates PASS) | 10,000                   |
| Cells rejected                        | 0                        |
| Pass rate                             | 100%                     |
| $\gamma$ range covered                | 14.134725 to 9878.654772 |

### 4.2 Measured Quantities

The dataset contains 33 fields per cell. The measured and directly derived quantities are reported here by their certified range — the minimum and maximum values actually observed across the 10,000 cells, with no averaging or statistical reduction applied.

| Quantity                              | Symbol          | Minimum observed        | Maximum observed        |
|---------------------------------------|-----------------|-------------------------|-------------------------|
| Breadth (Resolution B)                | $W_{\perp}$     | 0.007935                | 33.011479               |
| Normalized location (Resolution B)    | $x$             | 0.346159                | 0.652174                |
| Breadth-to-width ratio (Resolution B) | $\rho$          | 0.203165                | 16.651377               |
| Cell width                            | $h_n$           | 0.037698                | 6.887314                |
| Breadth discrepancy (A vs B)          | $\varepsilon_W$ | $1.042 \times 10^{-30}$ | $2.626 \times 10^{-15}$ |
| Location discrepancy (A vs B)         | $\varepsilon_x$ | $5.441 \times 10^{-27}$ | $8.957 \times 10^{-8}$  |

All cells remain within protocol tolerances. The maximum observed  $\varepsilon_W$  ( $2.626 \times 10^{-15}$ ) is roughly six orders of magnitude below the Gate 4 threshold of  $10^{-9}$ ; the maximum observed  $\varepsilon_x$  ( $8.957 \times 10^{-8}$ ) is roughly two orders of magnitude below the Gate 4 threshold of  $10^{-5}$ .

### 4.3 Breadth-to-Width Ratio: Frozen Definition

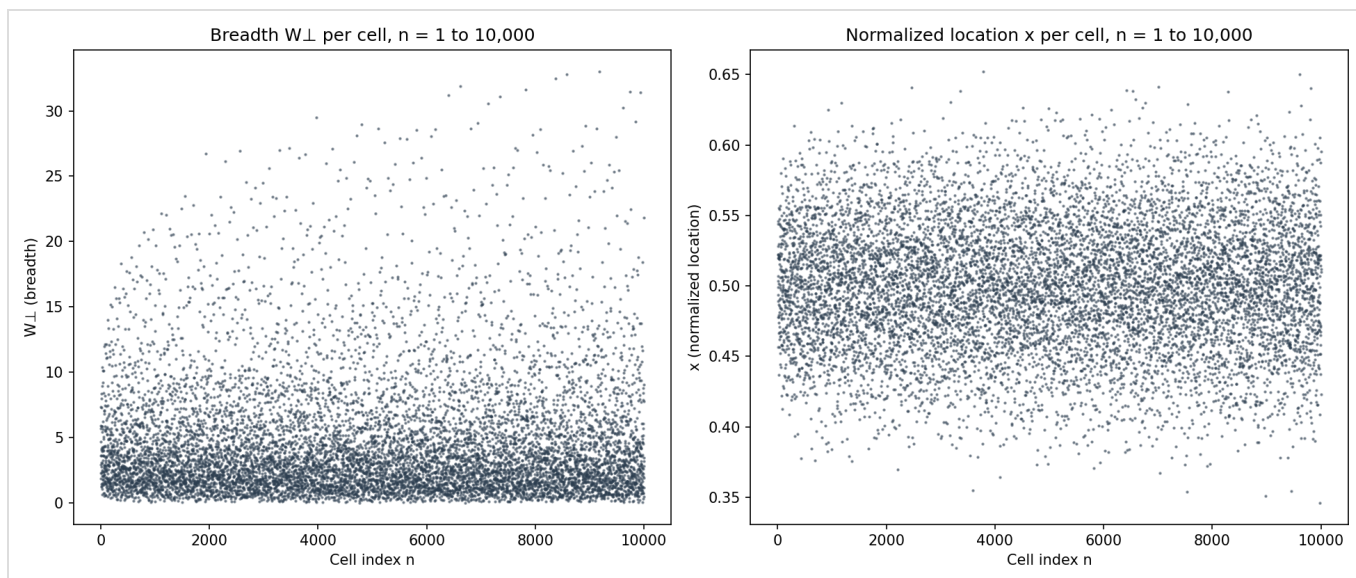
The breadth-to-width ratio is defined, before any measurement, as:

$$\rho_n = W_{n\perp} / h_n$$

This is a fixed ratio of two measured quantities. It is not fit, refit, or adjusted from the data at any stage, and it carries no additional assumption beyond the two quantities it divides.

### 4.4 Figure: Raw Cell-by-Cell Values

The two panels below plot the certified value of each of the 10,000 individual cells, in order by cell index  $n$ . No averaging, binning, or trend-fitting has been applied — each point is one certified cell.



## 5. Data Schema: Complete Field Documentation

Each row in the dataset corresponds to one certified Hardy cell. Each cell has 33 fields:

| Field Name                                                              | Description                                                                                                                                                   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| batch_k                                                                 | Batch number (1–10), for independent inspection in groups of 1,000 cells.                                                                                     |
| n                                                                       | Cell index (1–10,000).                                                                                                                                        |
| gamma_n, gamma_n1                                                       | Lower and upper zero boundaries of the cell (50 decimal places).                                                                                              |
| h_n                                                                     | Cell width ( $\text{gamma\_n1} - \text{gamma\_n}$ ).                                                                                                          |
| A_points, A_dps,<br>A_candidate_count, A_t_max,<br>A_W_perp, A_x        | Resolution A (64-point, 28 dps) measurement: grid size, precision, candidate maxima found, location and breadth of the selected maximum, normalized location. |
| B_points, B_dps,<br>B_candidate_count, B_t_max,<br>B_W_perp, B_rho, B_x | Resolution B (128-point, 40 dps) measurement: same quantities as Resolution A, plus $B_{\rho}$ , the breadth-to-width ratio at Resolution B.                  |
| epsilon_W, epsilon_x                                                    | Relative breadth discrepancy and absolute location                                                                                                            |

|                                                                |                                                                                                    |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|                                                                | discrepancy between Resolution A and Resolution B.                                                 |
| epsilon_W_limit,<br>epsilon_x_limit                            | Frozen Gate 4 thresholds ( $10^{-9}$ and $10^{-5}$ respectively).                                  |
| boundary_residual_left_50dps,<br>boundary_residual_right_50dps | Diagnostic: $ Z(t) $ magnitude at the cell's left and right zero boundaries, at 50 decimal places. |
| breadth_status                                                 | CERTIFIED for all cells in this dataset.                                                           |
| stationary_status                                              | UNRESOLVED for all cells in this dataset — stationary-root detection was not attempted.            |
| gate1_breadth_solver ...<br>gate5_raw_integrity                | Individual gate outcomes (PASS or FAIL) as specified in Section 2.                                 |
| all_five_gates                                                 | PASS only if every one of the five gates passed.                                                   |
| notes                                                          | Protocol version and measurement notes.                                                            |

## 6. Scope and Limitations

### 6.1 What This Dataset Measures

This dataset measures two independent quantities for 10,000 Hardy cells under a frozen protocol: the transverse breadth  $W_{\perp}$  of Hardy's Z-function, and the normalized location  $x$  of that breadth's interior maximum. The breadth-to-width ratio  $\rho$  is a fixed, deterministic function of the breadth and the cell width. The dataset does not attempt to prove or disprove any theory about Hardy cells. It records what the frozen protocol observed.

### 6.2 Independence Guarantees

This dataset makes the following independence claims: the protocol was frozen (September 2, 2026) before the certified run was acquired; each gate is designed to fail independently; no parameter was tuned, no threshold softened, no gate logic modified after data arrived; every cell that passed all five gates is included, with no cherry-picking of a "clean subset."

### 6.3 Limitations and Non-Claims

This dataset does not:

- Prove that the frozen gates are optimal or even good at discriminating cells. They apply a

deterministic rule.

- Establish or imply any stationary-point, curvature, or phase-space structure. That diagnostic was not attempted, and `stationary_status` is recorded as `UNRESOLVED` for every cell.
- Establish causation between any measured quantities and external phenomena. It records observations only.
- Validate any external model or theory. Validation happens when a theory makes a prediction and the data either matches or contradicts it.
- Claim that a 100% pass rate is "correct." It is a fact about the data, not a judgment about the protocol's sensitivity — the point is that the pass/fail decision was made before knowing the results.

## 6.4 Replicability and Reproducibility

This dataset can be replicated by any researcher who has access to a Hardy Z-function evaluator, implements the five gates exactly as specified in Section 2, obtains the same 10,000 zero boundaries from LMFDB, and runs each cell through the identical frozen protocol.

# 7. Significance and Interpretation

## 7.1 A Measurement Without Interpretation

This dataset is deliberately free of interpretation. It records: "When we apply this frozen protocol to these cells, these are the results." It does not say "this proves X" or "this refutes Y." It is a fact record, not an argument.

The significance lies in the certainty of the record: because the protocol was frozen, we can be confident that the results were not shaped by post-hoc adjustments or retroactive theory refinement.

## 7.2 The Validation Paradigm

This dataset enables forward prediction against frozen data. Any theory that makes specific predictions about Hardy cell breadth, location, or the breadth-to-width ratio can now be checked against this data without the theory's author having had any influence on how the data was collected or processed. The theory author cannot have unconsciously tuned the data collection to fit their theory, because the data collection was locked before the theory was tested against it.

# 8. Conclusion

We present 10,000 Hardy cells acquired and certified under Protocol v2, a frozen specification locked

before data collection on September 2, 2026. The dataset includes complete field documentation and convergence diagnostics. No retroactive adjustments were made. No gates were softened. No cells were retested.

The pass rate is 100%. This is a fact, not a judgment. The dataset stands as a record of what the frozen protocol observed. It is complete and requires no external framework to interpret. Its value lies in reproducibility, honesty, and the certainty that the results were not shaped by prior beliefs about what the answer should be.

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**Data Availability:** The complete dataset (10,000 cells, 33 fields each) is provided in the accompanying CSV file: `breathing_v2_cells_1_10000_master_CERTIFIED.csv`. Each row is a certified cell; each column corresponds to the fields documented in Section 5.

**Companion document:** *A Frozen Specification for the Measurement of Hardy-Z Breadth in Consecutive Zero Cells — Protocol v2*, the authoritative methods record for this dataset.

References: Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press. · Odlyzko, A. M. (1992). The  $10^{20}$ -th zero of the Riemann zeta function and 70 million of its neighbors. · Gourdon, P. (2004). Verification of the first trillion zeros of the Riemann zeta function. · LMFDB Collaboration. (2024). The L-functions and Modular Forms Database. <http://www.lmfdb.org>